

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-12	Team No- 18	Project Title- Automotive Mobile App for Vehicle Owner Self-Service
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1. Abstract

Modern vehicle ownership often involves challenges such as scheduling maintenance, tracking service history, and accessing roadside assistance. Traditionally, these tasks require manual intervention, phone calls, or physical visits to service centers, which can be time-consuming and inconvenient for owners. With the rapid growth of mobile technology and connected vehicles, there is a pressing need for a digital solution that empowers owners to manage their vehicles more efficiently.

This project proposes the development of an **Automotive Mobile App for Vehicle Owner Self-Service**, designed to streamline and centralize essential vehicle-related tasks. The app will enable users to book service appointments, monitor maintenance schedules, access digital service records, and request emergency roadside assistance—all from their smartphones. Additionally, features such as real-time notifications, warranty tracking, and integration with GPS for locating nearby service centers will enhance user convenience and reduce dependency on manual processes.

By providing a user-friendly interface and secure data management, the app aims to improve customer satisfaction, reduce operational bottlenecks for service providers, and promote proactive vehicle care. Ultimately, this solution bridges the gap between vehicle owners and service providers, offering a seamless, self-service ecosystem that aligns with the growing demand for digital transformation in the automotive industry.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

Mobile App Development Frameworks:

- **Flutter (Dart)** – Cross-platform mobile development for Android & iOS with a single codebase.
- **React Native (JavaScript/TypeScript)** – Alternative for building native-like apps with reusable components.

Backend & APIs:

- **Node.js with Express.js** – Lightweight backend for RESTful APIs.
- **Firebase** – For authentication, cloud storage, and real-time database.
- **Spring Boot (Java)** – Option for enterprise-grade backend services.

Databases:

- **MySQL** – Relational database for structured vehicle and user data.
- **MongoDB** – NoSQL database for flexible storage of service records and logs.

Third-Party Integrations:

- **Firebase Cloud Messaging** – Push notifications and SMS alerts.
- **Payment Gateway (Razorpay / Stripe / PayPal)** – For online service payments.

Development & Collaboration Tools:

- **GitHub** – Version control and collaboration.
- **Postman** – API testing and debugging.

Security & Authentication:

- **JWT** – Secure login and user session management.
- **SSL/TLS Certificates** – Encrypted communication for sensitive data.

Guide Name and Signature (remarks if any) _____